

CORMORANT STATION 63112, United KingdomWMO: **996920**Lat: **61.100N**Lon: **1.000E**Elev: **30**StdP: **100.97**Time Zone: **0.00 (EUW)**Period: **01-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.0	1.1	-7.3	2.0	2.1	-6.0	2.3	2.8	22.2	4.4	21.0	3.9	9.5	350	0.877

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	19.1	16.1	17.9	15.3	16.8	14.4	16.8	18.5	15.7	17.4	14.8	16.4	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.4	17.9	14.9	10.6	16.8	14.0	10.0	15.8	47.2	18.5	44.1	17.5	41.6	16.5	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.4	15.3	-1.3	22.5	1.4	2.7	-2.3	24.4	-3.1	26.0	-3.9	27.5	-4.9	29.5		
(5)			-2.3	18.0	1.1	2.3	-3.1	19.7	-3.8	21.1	-4.4	22.4	-5.2	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	6.1	5.8	6.3	7.7	9.5	11.8	13.9	14.5	12.8	10.1	8.5	6.5	
	DBStd	3.68	2.01	2.27	2.33	2.04	2.05	2.09	1.98	1.74	1.85	2.01	2.13	2.58	
	HDD10.0	651	122	118	114	72	34	5	0	0	2	23	52	109	
	HDD18.3	3237	381	351	372	320	276	196	139	118	167	256	296	366	
	CDD10.0	459	0	0	1	2	17	59	121	141	86	26	6	2	
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0	
	CDH23.3	1	0	0	0	0	0	0	1	1	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	8.0	9.9	9.5	8.7	7.5	6.8	6.1	6.3	6.4	7.4	8.5	9.3	9.7	
Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156	
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240	
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65	
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.7	11.9	13.2	16.4	20.0	21.4	21.2	18.3	15.2	13.6	12.5	
		MCWB	9.7	8.9	9.1	9.8	12.3	15.0	17.5	17.8	15.7	12.8	10.8	10.4	
	2%	DB	10.6	10.8	11.0	11.8	14.3	17.4	19.1	19.0	16.7	14.0	12.4	11.6	
		MCWB	8.5	8.5	8.6	9.0	10.9	13.8	16.1	16.6	14.2	11.6	10.0	9.7	
	5%	DB	9.6	9.6	10.1	10.9	13.1	15.8	17.8	17.8	15.8	13.2	11.5	10.6	
		MCWB	7.8	7.5	7.9	8.4	10.0	12.9	15.4	15.4	13.6	11.0	9.6	8.8	
	10%	DB	8.8	8.7	9.2	10.1	12.0	14.5	16.5	16.8	15.1	12.5	10.9	9.7	
		MCWB	7.2	6.8	7.2	7.9	9.5	12.2	14.5	14.6	13.2	10.5	9.1	8.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	9.7	9.6	10.6	12.7	15.5	17.8	18.3	16.1	13.5	11.5	10.8
			MCDB	11.2	10.9	11.3	12.4	15.7	19.2	20.4	19.8	18.2	14.8	12.7	12.0
2%		WB	9.0	8.9	8.9	9.5	11.4	14.3	16.6	17.2	14.8	12.3	10.7	10.1	
		MCDB	10.0	10.2	10.6	11.1	13.5	16.7	18.5	18.6	16.1	13.4	11.8	11.3	
5%		WB	8.2	7.9	8.2	8.8	10.6	13.2	15.7	16.0	14.1	11.6	10.1	9.2	
		MCDB	9.3	9.2	9.7	10.2	12.3	15.3	17.4	17.3	15.3	12.7	11.2	10.4	
10%		WB	7.4	7.2	7.5	8.3	10.0	12.5	14.8	14.9	13.5	10.8	9.5	8.4	
		MCDB	8.7	8.5	8.9	9.8	11.5	14.2	16.4	16.3	14.8	12.0	10.7	9.6	
Mean Daily Temperature Range		5% DB	MDBR	3.3	3.2	3.5	3.6	4.0	3.7	3.6	3.5	3.1	3.0	3.2	3.5
			MCDBR	4.4	4.4	4.3	4.8	5.5	5.9	5.6	5.1	4.1	3.9	3.8	4.3
	MCWBR		4.2	3.9	3.6	3.3	3.6	3.6	3.2	3.3	2.8	3.1	3.4	4.1	
	5% WB	MCDBR	4.3	4.1	4.1	4.4	5.0	5.2	5.1	4.9	3.8	3.6	3.7	4.2	
		MCWBR	4.4	4.0	3.7	3.4	3.4	3.4	3.2	3.6	2.7	3.3	3.8	4.2	
Clear-Sky Solar Irradiance	taub	0.335	0.355	0.368	0.380	0.370	0.363	0.380	0.369	0.365	0.346	0.321	0.385		
	taud	2.062	2.185	2.266	2.308	2.406	2.467	2.435	2.466	2.428	2.306	2.124	2.301		
	Ebn at Noon	345	595	739	810	853	866	838	819	748	608	360	217		
	Edh at Noon	45	72	92	106	103	99	100	90	79	63	40	29		
All-Sky Solar Radiation	RadAvg	0.19	0.68	1.81	3.51	4.90	4.99	4.53	3.63	2.20	1.04	0.30	0.12		
	RadStd	0.03	0.08	0.19	0.27	0.35	0.54	0.41	0.34	0.17	0.12	0.04	0.01		